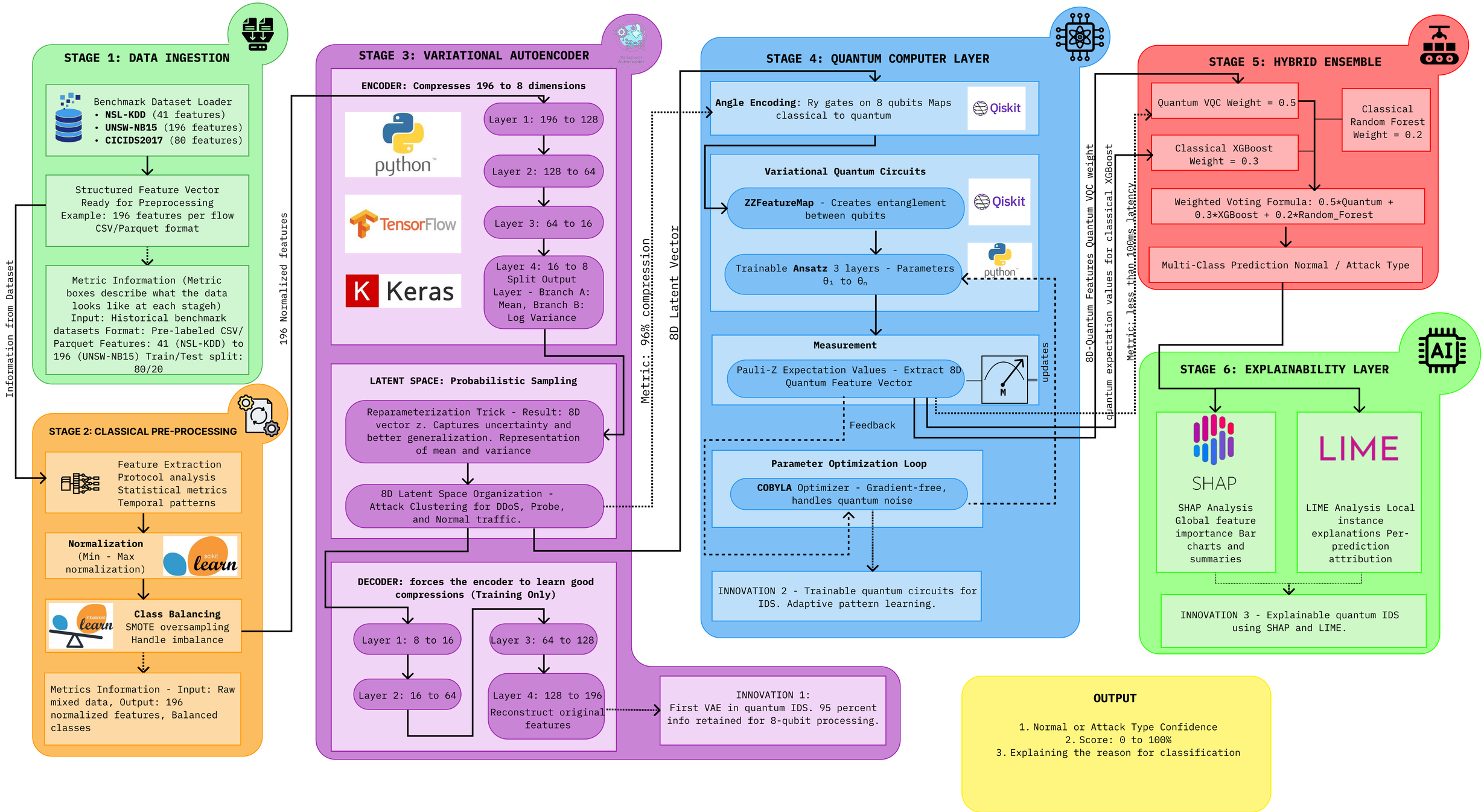
STAGE 1: DATA INGESTION
Benchmark Dataset Loader
• NSL-KDD (41 features)
• UNSW-NB15 (196 features)
• CICIDS2017 (80 features)
Structured Feature Vector Ready for Preprocessing Example: 196 features per flow CSV/Parquet format
Metric Information (Metric boxes describe what the data looks like at each stageh) Input: Historical benchmark datasets Format: Pre-labeled CSV/ Parquet Features: 41 (NSL-KDD) to 196 (UNSW-NB15) Train/Test split: 80/20
Information from Dataset
STAGE 2: CLASSICAL PRE-PROCESSING
Feature Extraction Protocol analysis Statistical metrics Temporal patterns
Normalization (Min - Max normalization)
Class Balancing SMOTE oversampling Handle imbalance
Metrics Information - Input: Raw mixed data, Output: 196 normalized features, Balanced classes
196 Normalized features
STAGE 3: VARIATIONAL AUTOENCODER
ENCODER: Compresses 196 to 8 dimensions
python
TensorFlow
Keras
Layer 1: 196 to 128
Layer 2: 128 to 64
Layer 3: 64 to 16
Layer 4: 16 to 8 Split Output Layer - Branch A: Mean, Branch B: Log Variance
LATENT SPACE: Probabilistic Sampling
Reparameterization Trick - Result: 8D vector z. Captures uncertainty and better generalization. Representation of mean and variance
8D Latent Space Organization - Attack Clustering for DDoS, Probe, and Normal traffic.
DECODER: forces the encoder to learn good compressions (Training Only)
Layer 1: 8 to 16
Layer 2: 16 to 64
Layer 3: 64 to 128
Layer 4: 128 to 196 Reconstruct original features
INNOVATION 1: First VAE in quantum IDS. 95 percent info retained for 8-qubit processing.
Metric: 96% compression
8D Latent Vector
STAGE 4: QUANTUM COMPUTER LAYER
Angle Encoding: Ry gates on 8 qubits Maps classical to quantum
Qiskit
Variational Quantum Circuits
ZZFeatureMap - Creates entanglement between qubits
Qiskit
Trainable Ansatz 3 layers - Parameters θ₁ to θₙ
python
Measurement
Pauli-Z Expectation Values - Extract 8D Quantum Feature Vector
M
updates
Feedback
Parameter Optimization Loop
COBYLA Optimizer - Gradient-free, handles quantum noise
INNOVATION 2 - Trainable quantum circuits for IDS. Adaptive pattern learning.
8D-Quantum Features Quantum VQC weight
quantum expectation values for classical XGBoost
Metric: less than 100ms latency
STAGE 5: HYBRID ENSEMBLE
Quantum VQC Weight = 0.5
Classical XGBoost Weight = 0.3
Classical Random Forest Weight = 0.2
Weighted Voting Formula: 0.5*Quantum + 0.3*XGBoost + 0.2*Random_Forest
Multi-Class Prediction Normal / Attack Type
STAGE 6: EXPLAINABILITY LAYER
SHAP
SHAP Analysis Global feature importance Bar charts and summaries
LIME
LIME Analysis Local instance explanations Per-prediction attribution
INNOVATION 3 - Explainable quantum IDS using SHAP and LIME.
OUTPUT
1. Normal or Attack Type Confidence
2. Score: 0 to 100%
3. Explaining the reason for classification